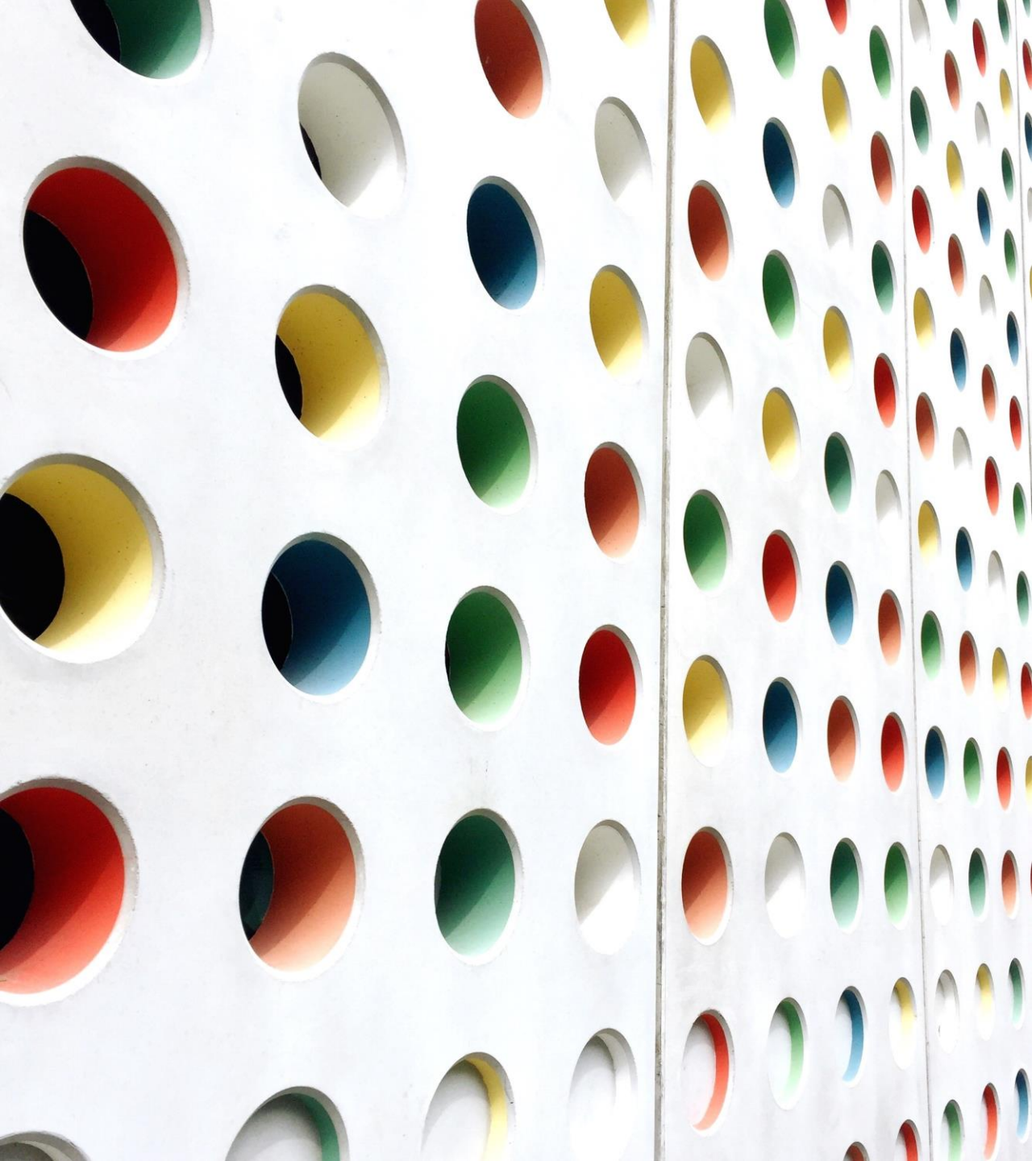


Tuberculosis

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- TB is an infection with mycobacterium tuberculosis.
- It is one of the top causes of all deaths worldwide.

- **Spread:** Respiratory droplet from person to person.

Specific risk factors for exposure:

- Health care workers
- Prisoners
- Homeless
- Shelter residents
- Chronically debilitated patients with weakened immune system
- lymphoma, leukemia, HIV, steroid use, organ transplant, chemotherapy, immunosuppressive.

Presentation

- Fever
- Cough> 3 weeks
- Sputum
- Weight loss
- Night fever
- Hemoptysis
- Any part of the body could be involved in 15-20% of cases. (lymph node, meninges, gastrointestinal system, genitourinary system, etc)

Signs

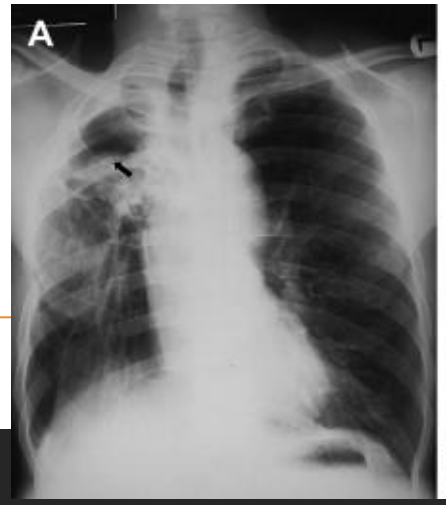
Sometimes difficult to distinguish from pneumonia

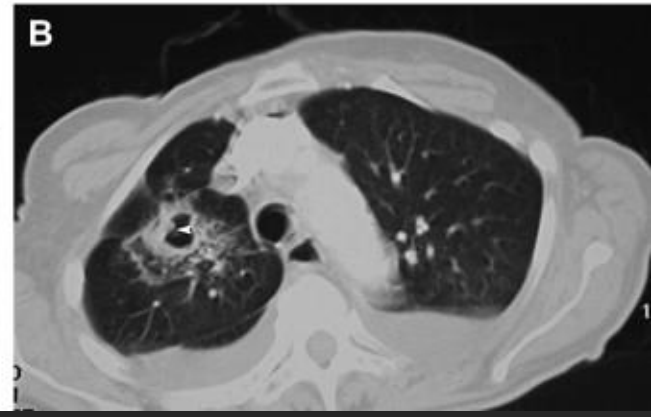
Commonly presented with apical fibrocavitary lesion.

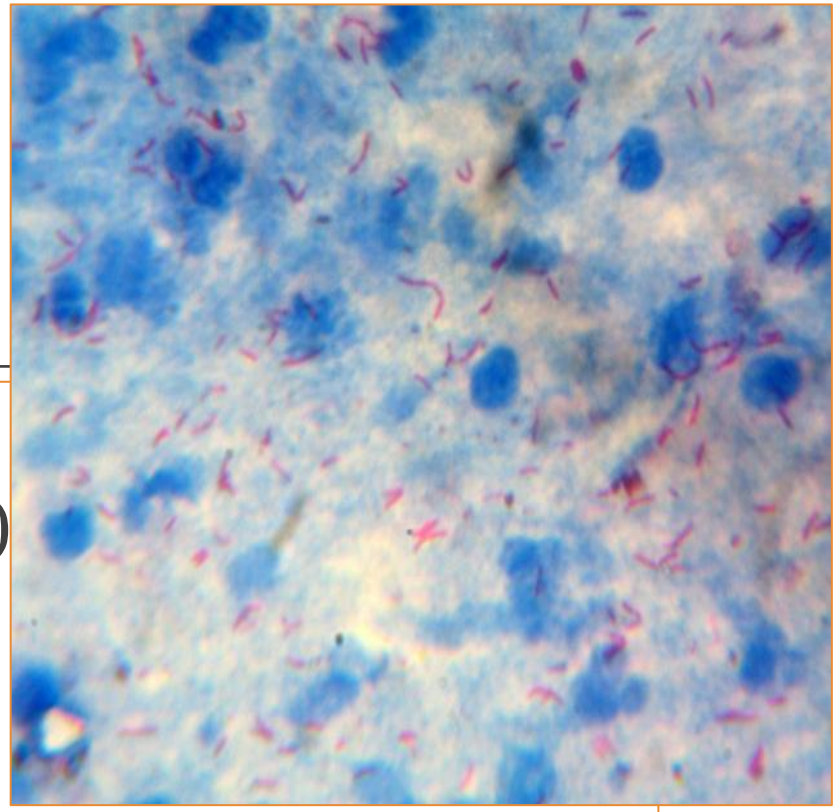
- Restricted mobility and retraction on the affected side.
- Decrease or heterogenous TVF & trachea shifted to same side of the lesion.
- Impaired note on affected side.
- Bronchial breathing and crepitations on affected side.

Diagnosis

- clinical
- Chest x ray
  - Apical involvement with infiltrate and sometime cavitation is the most common finding.
  - Adenopathy, effusion and calcified nodules are associated finding.







- Sputum examination :
  - direct smear with Zeihl neelsen stain (AFB)
    - ✓(25% is false negative)
    - ✓Trigger the treatment
  - culture on Lowenstein Jensen media.
    - ✓ Culture considered a gold standard but take 6 weeks to grow.
    - ✓Culture is important for sensitivity.

- Thoracocentesis: if there is pleural effusion.
- Pleural fluid adenosine deaminase could be helpful.
- Pleural biopsy (75% sensitivity)
- Gastric aspirate in children
- Lymph node biopsy
- Fine needle aspiration (FNA) of specific extrapulmonary involved organ.
  - Biopsy could reveal caseating necrosis.

- Lumbar puncture in meningitis
- Gene expert: to diagnose TB and rifampicin resistance in two hours.
- Diagnosis of latent TB in which absence of symptoms and normal x ray: tuberclin test , interferone gamma release assay (IGRA).

Treatment

Drug	Dose	Side effects
Isoniazide (INH) or (H)	5 mg/Kg	Peripheral neuritis, hepatitis, hypersensitivity
Rifampicin (Rif) or (R)	10mg /Kg	Orange urine Flu like illness Hepatitis Hypersensitivity Blood dyscriasis
Ethambutol (ETB) or (E)	25mg/Kg for two months, then 15mg/Kg	Retrobulbar neuritis
Pyrazinamide (PZA) or (Z)	<50Kg,1.5g 50-74Kg, 2g >75Kg, 2.5g	Hepatotoxicity, Hyperuricaemia
Streptomycin (S)	20mg/Kg (max 1g)	Ototoxicity (vestibular) Nephrotoxicity Hypersensitivity

New case :

- either pulmonary or extrapulmonary 6 month regimen

➢2 months Isoniazide (INH), Rifampin (Rif), Ethambutol (ETB), Pyrazinamide (PZA)/ 4 months Isoniazide (INH) and Rifampicin (Rif).

➢Forms of TB that must be treated for more than six months: TB meningitis (12 months), TB in pregnancy and osteomyelitis (9 months)

➢HIV positive (6 or 9 months)

➢Use of INH must be combined with vit B6 to prevent peripheral neuropathy.

Previously treated smear positive patients

2 SRHZE/1 RHZE/5 RHE (8 month regimen)

Sensitivity pattern is recommended

Chronic smear positive patient& MDR-TB & X TB

- Treated in hospital
- Sensitivity pattern is recommended to give special treatment regimen.

Indications of steroid in tuberculosis

- In very ill patient.
- To control drug hypersensitivity.
- In tuberculosis of serous sacs pericarditis, peritonitis and pleural effusion.
- In tuberculous meningitis.
- Genitourinary tuberculosis.

Diagnosis and treatment of latent TB

- Latent TB mean when a person is infected with Mycobacterium tuberculosis, but does not have a disease.
- Patients with a clinically significant result on tuberculin skin testing or a positive interferon-gamma release assay (IGRA) should receive a course of therapy for latent TB, once active infection and disease are ruled out.
- Isoniazid 300 mg - Daily for 9 months or
- Rifampin 600 mg - Daily for 4 months.

MCQs

1. Which one of the following is contraindicated in tuberculous patient suffered from chronic renal failure?

   a. Rifampicin.
   b. Isoniazide
   c. Pyrazinamide
   d. Ethambutol
   e. Streptomycin

2. A 25 year old patient presented to ER with fever, cough and hemoptysis of one month duration, which of the following investigations is recommended?

   a. Pulmonary function test.
   b. Sputum analysis for tuberculosis (AFB)
   c. Tuberclin test
   d. Interferone gamma release assay (IGRA)
   e. Bronchoscopy

3. Which one of the following is not a sign of right fibro-cavitary lesion?

a. Retraction on right side
b. Trachea shifted to left.
c. Impaired note on right side
d. Bronchial breathing
e. Crepitations

4. A 48 year old male patient suffer from chronic liver disease develop pulmonary tuberculosis, which one of the following antituberculous drugs should be avoided?

a. Streptomycin
b. Azithromyin
c. Rifampicin
d. Ethambutol
e. Flouroquinolone

5. Which one of the following is an indication of steroid in tuberculosis?

   a. Pulmonary tuberculosis.
   b. Tuberculous pleural effusion
   c. Liver tuberuloma
   d. Tuberculous cervical lymphadenopathy.
   e. Tuberculous enteritis

6. Which test is helpful for screening of latent TB?

   a. Sputum direct smear for AFB
   b. X ray
   c. Bronchoalveolar lavage
   d. IGRA
   e. Gene expert

7. A 25 female patient develops pulmonary tuberculosis, For how long she must receive anti TB treatment?
   a. 2 months
   b. 6 months
   c. 8 months
   d. 12 months
   e. 16 months

8. Which one of the following is the most common presentation?
   a. Pulmonary collapse
   b. Fibrocavitary lesion
   c. Pleural effusion
   d. Pulmonary nodule
   e. Consolidation

9. Which one of the following is a side effect from pyrazinamide?
   a. Peripheral neuropathy
   b. Impairment of renal function
   c. Retrobulbar neuritis
   d. Ototoxicity
   e. Hyperuricemia

10. A 31 year old male newly diagnosed with pulmonary TB, which one of the following regimens is suitable for her?

a. Isoniazide 300 mg daily for 9 months
b. Rifampicin 600mg daily for 6 months
c. 2 months Isoniazide (INH), Rifampin (Rif), Ethambutol (ETB), Pyrazinamide (PZA) followed by 4 months Isoniazide (INH) and Rifampicin (Rif).
d. 4 months Isoniazide (INH), Rifampin (Rif), Ethambutol (ETB), Pyrazinamide (PZA) followed by 2 months Isoniazide (INH) and Rifampicin (Rif).
e. 4 months Isoniazide (INH), Rifampin (Rif), Ethambutol (ETB), Pyrazinamide (PZA) followed by 4 months Isoniazide (INH) and Rifampicin (Rif).


Thank you